

Evaluating XAI Support From A Hierarchical Reinforcement Learning Policy in Human-Agent Collaboration

Extended Abstract

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ABSTRACT

Explainable AI (XAI) research in human-agent collaboration has relied on hand-crafted policies in custom environments, limiting generalizability to state-of-the-art teaming benchmarks. We provide the first systematic evaluation of XAI support generated from a learned, intrinsically explainable policy in an established benchmark environment. We generate real-time explanations from hierarchical subtask selections via a trigger-based delivery system, comparing text versus audio modalities in a between-subjects experiment with gaming-experienced participants. While we found no statistically significant performance benefits, preliminary patterns suggest expertise may moderate explanation utility. This work establishes a methodological foundation for evaluating intrinsically explainable reinforcement learning in collaborative contexts, along with the first comparison of explanation modality effects in real-time human-agent teaming.

KEYWORDS

Human-Agent Teaming; Hierarchical Reinforcement Learning; XAI

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1 INTRODUCTION

Research in explainable AI (XAI) and human-agent teaming (HAT) has remained largely disconnected: while XAI techniques have been developed for various AI systems [9], systematic evaluation of XAI in established HAT benchmarks remains limited [12]. HAT research in benchmark environments like Overcooked-AI [3] has prioritized performance optimization over transparency [6, 10], while the few works applying XAI to collaborative contexts have relied on custom environments [7] or non-deployable techniques [13].

We bridge this gap by leveraging the Hierarchical Ad Hoc Agents (HA²) architecture [1], which achieves state-of-the-art performance through intrinsically explainable hierarchical policies. HA²'s Manager selects human-interpretable subtasks while Workers execute

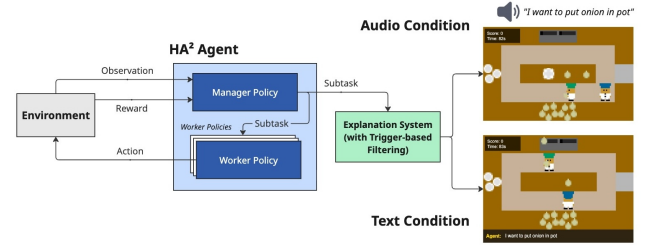


Figure 1: System architecture and experimental conditions. The Manager’s subtask selections flow to both the Worker (for action execution) and the Explanation System with trigger-based filtering. Identical explanation content (“I want to put onion in pot”) is delivered via audio speech synthesis (top right) or text display (bottom right, black bar).

primitive actions, providing natural explanation content without post-hoc training [11], yet this explanatory potential remains unexplored.

We generate real-time explanations from HA²'s subtask selections during collaborative Overcooked-AI tasks. Since pilot testing revealed that continuous explanation display overwhelmed users, we developed a trigger-based delivery system generating explanations only at coordination-relevant moments. We compare text versus audio modalities, hypothesizing audio reduces cognitive interference in visually demanding tasks in a between-subjects experiment (n=38).

To our knowledge, we provide the first systematic evaluation of XAI support generated from a learned policy in an established benchmark, comparing different modalities while keeping content identical as well as providing a baseline trigger-based delivery system for managing explanation frequency for real-time collaboration. While our exploratory study found no statistically significant performance benefits from explanations, preliminary patterns suggest the peak performance of gaming experts may not benefit from simple status-based explanations.

2 METHODOLOGY

Agent Implementation: We use HA² [1] with the original paper’s Worker policies and a Manager trained via PPO self-play on Overcooked-AI’s Counter-Circuit layout. The Manager selects from twelve domain-specific subtasks (e.g., “pick up onion from dispenser”, “place onion in pot”) each timestep, with Workers executing corresponding primitive actions.



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Trigger-Based Explanation System: Since the Manager updates subtask selections every timestep, pilot studies revealed that directly exposing these choices created excessive and confusing communication. To remedy this, we developed a priority-based trigger system generating explanations only at coordination-relevant moments: (1) *Blocking* – when the human occupies the agent’s path, (2) *Critical Path* – when the agent selects task-essential subtasks, (3) *Distance Threshold* – when the agent travels predetermined distance since last explanation, and (4) *Subtask Change* – when the Manager switches to new subtask. Each trigger maps to natural language templates (e.g., “Moving around you to [subtask]”). A six-second cooldown prevents excessive communication; text explanations are displayed on the game interface for 4 seconds, while audio explanations are generated via browser-native speech synthesis.

Experimental Design: Between-subjects design with three conditions: Control (no explanations), Text (visual display, 4-second duration), Audio (identical content via browser-native speech synthesis). Following recommendations established in [8], participants completed four 80-second Counter-Circuit sessions with NASA-TLX assessments [4] after each session and Godspeed Questionnaire [2] and Human-Agent Fluency Assessment [5] post-experiment. We implemented the experiment via a web-based platform supporting remote participation.

Participants: Following IRB approval, we recruited 41 participants via convenience sampling from the academic community, yielding $n=38$ after excluding participants with $<80\%$ instructional check accuracy (ages 19-43, $M=25.2$, $SD=4.5$; $\approx 66\%$ male; Control $n=14$, Text $n=14$, Audio $n=10$). Post-hoc analysis revealed high gaming familiarity ($M=8.1/10$, $SD=2.1$; 92% scoring $\geq 6/10$). We targeted $n=30$ based on prior work [7] expecting large effects, but sensitivity analysis revealed we were underpowered for the medium effects we observed ($H1: d=0.51$; $H2: d=0.14$).

3 PRELIMINARY FINDINGS

Performance: Gaming-experienced participants trended toward higher scores *without* explanations (Control: $M=125.00$, $SD=18.19$ vs. Explanations: $M=114.17$, $SD=22.59$; $t(36)=-1.53$, $p=.136$, $d=-0.51$), though this did not reach significance. Modality comparison showed equivalent performance (Text vs. Audio: $t(22)=0.33$, $p=.745$, $d=0.14$); NASA-TLX cognitive workload results revealed no significant differences across any dimension after Bonferroni correction, and subjective collaboration measures (Godspeed, Human-Agent Fluency) showed no significant effects (all $p>0.05$).

Learning Rates: Despite equivalent peak performance, learning trajectories revealed a monotonic pattern (Figure 2). Audio participants improved fastest ($M=11.20$ points/session), followed by Text ($M=8.00$), then Control ($M=5.39$) – a $2.1\times$ difference between Audio and Control. While not reaching significance ($F(2,34)=1.51$, $p=.236$), this medium effect ($d=0.50$, $p=.158$) suggests explanations may accelerate adaptation rather than improve execution. This pattern of equivalent final performance despite different learning rates indicates that transparency may facilitate mental model development without necessarily translating to peak performance advantages, particularly among expert users who can rapidly identify efficient strategies.

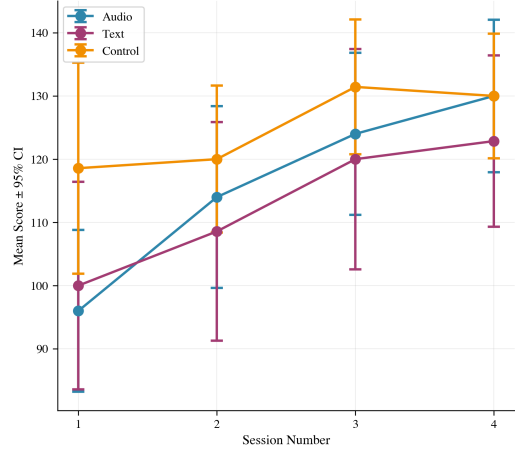


Figure 2: Learning curves by condition showing equivalent peak performance but differential adaptation rates. Audio explanations produced the steepest learning trajectory despite no final performance advantage.

Modality Salience: In qualitative feedback, multiple text participants reported not noticing explanations until later sessions, with no equivalence in audio participant responses. This salience difference may explain differential learning rates despite equivalent cognitive load and peak performance. Audio participants also trended toward rating the agent as less anthropomorphic ($p=.081$) and trustworthy ($p=.100$), suggesting that more transparent decision-making visible may expose the agent’s artificial nature – particularly when the Manager policy changed subtasks mid-execution, creating mismatches between stated intentions and observed actions.

4 CONCLUSIONS

This exploratory work demonstrates that intrinsically explainable RL architectures can generate authentic real-time explanations in benchmark environments. Our trigger-based delivery system addresses the practical challenge of managing explanation frequency in fast-paced tasks, while our modality comparison reveals that audio ensures greater salience than text – though neither improved peak performance for our gaming-experienced sample. The preliminary pattern of faster adaptation with audio explanations despite equivalent final scores suggests XAI may facilitate mental model development rather than direct execution benefits, particularly among expert users. Adequately-powered further studies with more diverse populations and multiple layouts are needed to confirm these exploratory findings.

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